Team-02

S. No.	Name	Student ID	Email ID
1	Shubham Kumar	2024HB21026	2024hb21026@wilp.bits-pilani.ac.in
2	Priyanka Nandakumar Pawar	2024HB21031	2024hb21031@wilp.bits-pilani.ac.in
3	Ravi DN	2025MN21152	2025mn21152@wilp.bits-pilani.ac.in
4	Cindhu K E	2024HB21027	2024hb21027@wilp.bits-pilani.ac.in

Experiential Learning – Phase 1 Submission

Project Title

E-Commerce Review Intelligence System Using NLP and Transformer Models for Customer Sentiment, Topic Extraction, and Business Insights

1. Business Problem Brief

Modern e-commerce platforms receive thousands of customer reviews containing valuable information about product quality, features, pricing, delivery, customer service, and overall customer satisfaction. However, these reviews are predominantly unstructured text, making it difficult for business teams to manually analyze customer feedback at a scale.

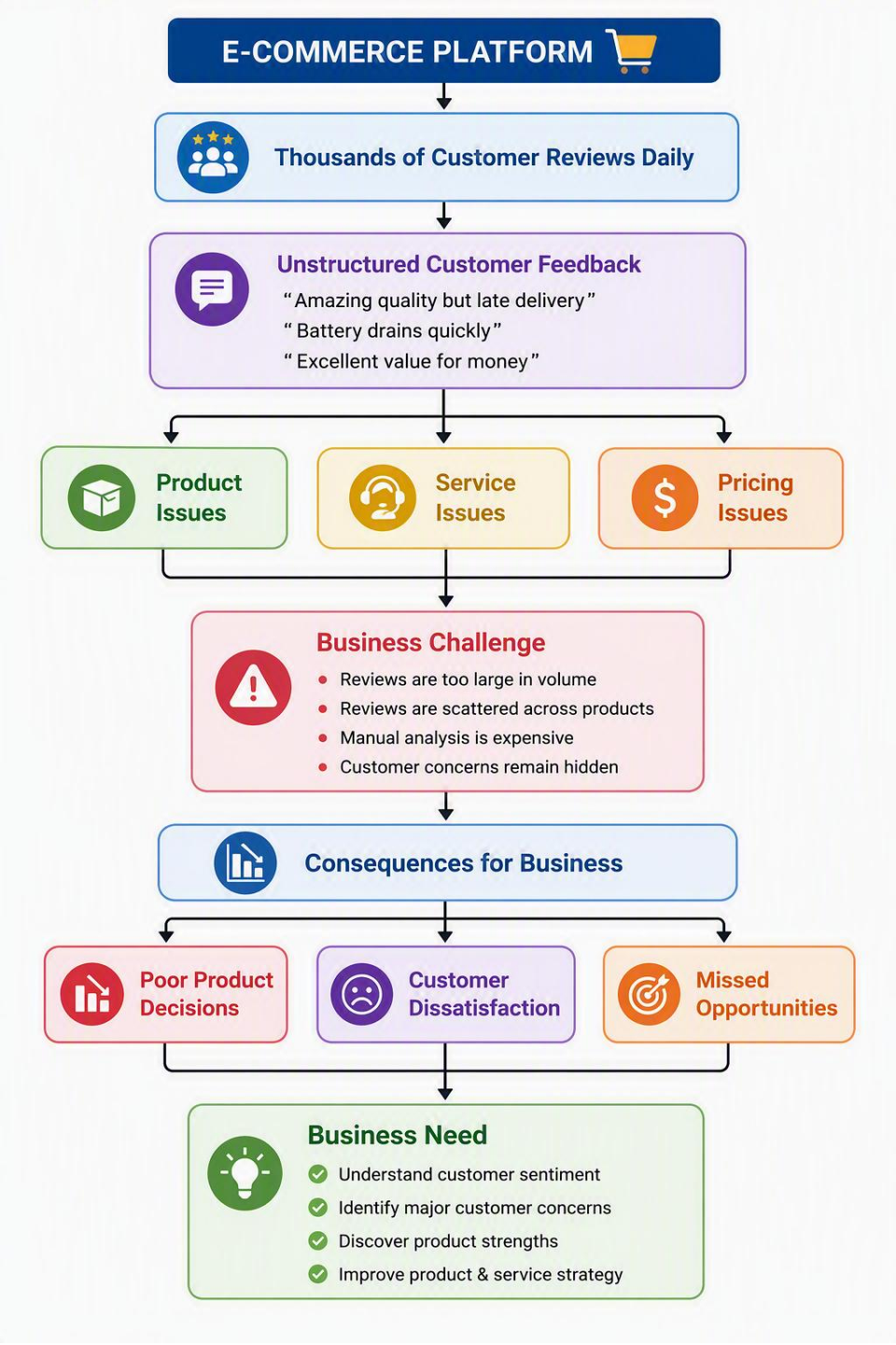
For example:

“The product quality is amazing, but delivery was delayed.”

“Good phone, but the battery drains quickly.”

“Excellent value for money.”

When large volumes of customer reviews are not systematically analyzed, customer dissatisfaction trends may remain hidden, product issues may be identified late, and marketing and product teams may miss opportunities to improve product features, pricing, and customer experience.



This project proposes an **E-Commerce Review Intelligence System** that automatically analyzes customer reviews to identify overall customer sentiment and major topics being discussed. The system will help answer the following business questions:

- **What** are customers saying about the product?
- **Why** are customers satisfied or dissatisfied?
- **How** can these insights support improvements in product, pricing, marketing, and service?

The proposed solution can help organizations identify product strengths and weaknesses. For example, customers may provide positive feedback regarding battery performance while expressing negative sentiment about camera quality or product design. Such insights can support business decisions related to product improvement, pricing, promotional offers, target customer segmentation, marketing strategy, and customer service.

Where suitable competitor review data is available, the approach may also support a comparison of customer perceptions regarding product features, value for money, and service offerings. The same framework can additionally support market analysis before launching a new product by identifying customer expectations and competitor strengths and weaknesses.

2. Proposed NLP Techniques

The project will use the following NLP techniques:

a) Text Preprocessing and Tokenization:

Customer review text will be prepared for analysis by handling missing values, duplicate records, unnecessary text, and formatting inconsistencies. The DistilBERT tokenizer will be used to convert the review text into a format suitable for the Transformer model.

b) Sentiment Analysis – Primary NLP Technique:

The primary objective is to classify customer reviews into:

- **Positive**
- **Neutral**
- **Negative**

A Transformer-based **DistilBERT** model will be used as the primary model because it can understand the contextual meaning of customer reviews while requiring relatively lower computational resources.

A traditional machine learning model using **TF-IDF and Logistic Regression** will be used as a baseline for comparison.

c) Topic Extraction – Secondary NLP Technique:

Topic extraction will be used to identify major themes discussed in customer reviews, such as:

- Product Quality
- Battery
- Camera
- Screen

- Price
- Delivery
- Product Design
- Customer Service
- Warranty

BERTopic, or a simpler topic modeling/clustering technique depending on implementation feasibility, will be used to identify major customer discussion topics.

The final analysis will combine **sentiment and topic information** to generate business insights. For example:

Positive sentiment: Battery performance and value for money

Neutral sentiment: Screen size

Negative sentiment: Camera quality and product design

This will help business teams identify key customer strengths, pain points, and improvement opportunities.

3. Dataset Description

The project will use a publicly available **e-commerce customer review dataset** from Kaggle or another open-source repository. The dataset will primarily contain unstructured customer review text and product ratings.

Typical data fields may include:

Review Text	Rating
Excellent product	5
Battery performance is poor	2
Average experience	3

Where explicit sentiment labels are not available, ratings will be converted into sentiment categories as follows:

- **1–2 → Negative**
- **3 → Neutral**
- **4–5 → Positive**

The dataset will be divided into training, validation, and test datasets for model development and evaluation.

The proposed workflow is:

Customer Reviews → Preprocessing and Tokenization → TF-IDF + Logistic Regression Baseline / DistilBERT Sentiment Classification → Topic Extraction → Business Insights

Expected Outcome

The expected outcome is a proof-of-concept system capable of automatically analyzing large volumes of customer reviews and generating structured insights on customer sentiment and major discussion topics. The solution is expected to help business teams better understand customer satisfaction, identify product-related issues, and support informed decisions related to product improvement, pricing, marketing, and customer service.